MOBILE DEVELOPMENT PROJECT REPORT

Academic Mobile Application · Problem Statement & Design Analysis

PROJECT TEAM

SM	SA	RG	MM	AS	AS
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1. PROBLEM STATEMENT

◆ Overview

Many universities rely on fragmented digital systems, forcing students, instructors, and teaching assistants to access different portals for schedules, grades, and announcements. This fragmentation increases cognitive load, causes missed deadlines, and degrades the overall academic experience.

◆ Target Users

The application serves three user groups: Students, who need consolidated access to schedules, grades, and registration; Instructors, who manage course content and communicate updates; and Teaching Assistants, who coordinate between students and faculty.

◆ User Pain Points

Students navigate multiple systems for exams, attendance, and fees. Instructors lack a unified channel for announcements and course management. Teaching assistants have inconsistent access to tools shared between both roles. All three groups suffer from delayed communication and administrative inefficiency.

2. PROPOSED SOLUTION

◆ What the App Does

The application is a role-sensitive mobile platform that consolidates academic management for all three user types within a single interface. A shared landing page routes each user type to a dedicated authentication flow, after which they access a personalised home screen and relevant service panels.

◆ Key Features

Student	Instructor	Teaching Assistant
· Class Schedule	· Class Schedule	· Class Schedule
· Exam Schedule	· Exam Schedule	· Exam Schedule
· Attendance	· Attendance Tracking	· Attendance View
· Courses & Registration	· Course Management	· Course Access
· Grades & Tuition Fees	· News & Announcements	· Home & Events
· Advisor Access	· Community Tools	· Profile & Settings

3. USER PERSONAS

◆ Student Persona

LH

Layla Hassan

3rd-year Student · Computer Science

Goal	Access her exam timetable, grades, and attendance in one place without using multiple portals.
Pain Points	Misses notifications buried in email. Finds the web portal slow on mobile. Cannot easily track attendance percentage.

◆ Instructor Persona

DK Dr. Karim Mansour *Associate Professor · Business Administration*		
	Goal	Push announcements to students, track attendance across sections, and manage course content efficiently.
	Pain Points	Wastes time updating multiple platforms. Lacks real-time attendance visibility. No unified notification channel.

4. USER STORIES

User	I want to…	So that…
Student	view my class schedule in one place	I can plan my week and avoid conflicts
Student	check my attendance percentage per course	I avoid academic penalties before it is too late
Student	access my exam timetable from the app	I prepare for assessments without visiting another portal
Student	complete course registration from my phone	I secure my place in required courses on time
Instructor	publish news and updates to my students	important information reaches them promptly
Instructor	monitor attendance across my course sections	I identify absenteeism patterns early
Instructor	manage my courses and exam schedules	I reduce administrative load and ensure accuracy
TA	access class schedules and course info	I fulfil my support responsibilities effectively
TA	view student attendance for my assigned sections	I flag concerns to the instructor in time

5. WIREFRAME OVERVIEW

◆ App Structure

The application is anchored by a central Landing Page that presents three role-selection options. Each role leads to a dedicated Login screen and, for new users, a Sign-Up screen with a Forgot Password recovery flow. Once authenticated, users enter a role-specific Home screen from which all core functionality is accessible through a persistent bottom navigation bar.

◆ Navigation Flows

Student flow: Home (Events, Announcements, Notices) → Services (Class Schedule, Exam Schedule, Attendance, Courses, Registration, Tuition Fees) → Community → Advisor → Settings. Instructor flow: Home (Events, News) → Services (Schedule, Exam, Attendance, Courses) → Community → Settings. TA flow mirrors the instructor structure. All flows share Settings (Dark Mode, Notifications, Edit Info, Logout). Two universal screens — 404 Error and Cookie Policy — apply across all roles.

6. DESIGN DECISIONS

◆ Screen Selection Rationale

Every screen corresponds directly to an identified user need. The Tuition Fees screen, for example, eliminates the need to visit a separate billing portal. Separate navigational trees per user type prevent confusion from inaccessible menu items, and the shallow information architecture minimises the number of taps to reach any given screen.

◆ Design Choices Supporting User Needs

A persistent bottom navigation bar provides fluid section transitions throughout all role-specific flows. Dark mode and notification controls give users agency over their experience. The blue-dominant palette conveys trust and academic credibility. Card-based layouts in the Services sections support fast, scannable information retrieval on small screens.

CONCLUSION

This report has presented the problem statement, proposed solution, user personas, user stories, wireframe structure, and design rationale for a mobile academic management application serving Students, Instructors, and Teaching Assistants. By consolidating fragmented institutional systems into a unified, role-sensitive mobile platform, the application addresses documented pain points and provides a more efficient and coherent academic experience. The design decisions outlined herein serve as a foundation for subsequent high-fidelity prototyping and usability testing phases.